



To the Editors of
Estuarine, Coastal and Shelf Science

Dear Editors,

We submit for your consideration the manuscript “**Transport validation of a wave-coupled lagoon model: interacting effects of seagrass roughness and bed mobility**”, by Cicero Martins Jr., Giuseppe Ciraolo and Mauro De Marchis, as an original research paper.

The corresponding author is MSc. Cicero Martins Jr.

Coastal lagoon models are commonly validated against water level and then used to describe transport. This manuscript asks whether the first licenses the second. We built six configurations of a three-dimensional Delft3D Flexible Mesh model coupled online with SWAN, differing in wave coupling, bed mobility and whether seagrass resistance is uniform or distributed from a satellite classification, and scored each against three tide gauges and 35 GPS surface drifters over the same nine days.

The configurations are indistinguishable in water level. Their anomaly RMSE differs by six millimetres or less at any station, which is within measurement uncertainty, so a calibration conducted on water level alone would have no basis on which to prefer any of them. Over the same simulations their Liu–Weisberg skill spans 0.505 to 0.655 and their endpoint separation 334 to 661 m.

What separates them is an interaction rather than any single process. Distributed seagrass roughness and bed mobility each change Lagrangian skill by an amount indistinguishable from zero when applied alone, and together they raise it by 0.13 to 0.15. Decomposing the trajectory error into a magnitude part and a direction part shows why: bed mobility restores the distance travelled, from 0.79 to 0.94 of observed, while distributed roughness reduces the mean heading error from 12.5 to 8.4 degrees and does so only once the bed can move. We report this as a narrowing rather than a complete mechanism, and we state the three candidate explanations we tested and had to reject.

We believe the work suits *Estuarine, Coastal and Shelf Science* on two grounds. It is a study of brackish water and lagoon phenomena and of coastal sedimentary processes, both within the journal’s stated scope. And although the model is built for one basin, the question it answers is not local: the finding concerns what a class of validation data can and cannot certify about a model, and the magnitude-direction decomposition we introduce is applicable wherever drifter observations and a circulation model exist together. The prior study closest to our wave treatment, Carniello et al. (2011), appeared in this journal.

We also draw your attention to two results we report against our own interest. Wave coupling, which we introduce to this system for the first time, does not improve Lagrangian prediction over this window. And the configuration that best reproduces observed transport depends on a morphodynamic parameter we have not been able to constrain against a repeated bathymetric survey, which we state explicitly as a limitation rather than leaving to a reader to discover.

The corresponding author affirms that:

- All authors have directly participated in the planning, execution, or analysis of this study;



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- All authors have read and approved the version submitted;
- The contents of this manuscript have not been copyrighted or published previously;
- The contents of this manuscript are not now under consideration for publication elsewhere.

We hope the manuscript is suitable for review, and we would be glad to provide any further clarification.

Sincerely,

Cicero Martins Jr.

on behalf of all co-authors

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